Traffic Monitoring System

https://github.com/aiyshwary/Traffic-Monitoring-System

PHP MySQL JavaScript HTML CSS XAMPP

OVERVIEW

A PHP and SQL-based web application that supports traffic controllers with real-time violator tracking, periodic traffic status notifications, daily reports, and transport department recommendations for infrastructure improvements.

IMPACT

Developed a multi-role Traffic Monitoring System web application with separate portals for admins, officers, and citizens, enabling automated violator tracking, fine management, road signalling status, and daily traffic reporting.

TECHNICAL WALKTHROUGH

(TMS)

Date: March 8, 2026
Built with PHP, MySQL, Apache (MAMP) | Bootstrap 4 Frontend

Project Overview
The Traffic Monitoring System (TMS) is a web-based application designed to streamline traffic management for both authorities (Officers) and the public (Citizens). It provides tools for monitoring road conditions, managing traffic signal timings, and handling traffic violation fines - including online payment. The system is built using PHP for server-side logic, MySQL as the relational database, Apache as the web server (via MAMP), and Bootstrap 4 for a responsive frontend UI.

Technology Stack

Layer Technology / Library
Backend Language PHP 8.3 (server-side scripting)

Database MySQL 8

Web Server Apache

Frontend Framework Bootstrap 4

JavaScript Libraries jQuery 2/3, Owl Carousel, ScrollReveal
CSS Libraries Font Awesome 4.7, Animate.css, Perfect Scrollbar
UI Utilities Select2, Tilt.js, Lightbox

Session Management PHP Native Sessions

User Roles

The system has three distinct user roles, each identified by a file prefix convention

> **i)** Prefix Role Access Level

ii) a Admin / Public Landing page, login/register chooser. No authentication required. c Citizen Registered vehicle owners. Can view own fines, pay fines, view signals and roads. o Officer Traffic authority personnel. Can manage fines, edit signal timings, view road data.

iii) File Structure & Purpose

iv) 4.1 Shared / Core Files

v) File Purpose

vi) aconnection.php MySQL DB connection - included by all PHP pages. Holds host, user, pass, dbname. logout.php Destroys the PHP session and redirects the user back to ahome.php. database.sql Full MySQL schema - creates all 5 tables and inserts sample road/signal data. 4.2 Public / Admin Pages (prefix: a) ahome.php Public homepage. Slider, About, Features, Team, Contact sections. alioption.php Login chooser page - two buttons: Citizen Login or Officer Login. asuoption.php Register chooser page - two buttons: Citizen Signup or Officer Signup. alogout.php Legacy logout file kept for compatibility.

vii) 4.3 Citizen Pages (prefix: c)

viii) csignup.php Registration form. Inserts into 'citizen' table AND auto-creates a fine record. clogin.php Login form. Validates against 'citizen' table, sets $_SESSION['email']. cfunctions.php check_login() function - validates session, redirects to clogin.php if absent. chome.php Citizen dashboard with slider linking to all citizen features. cfine.php Shows own fine records (amount, status). Pay Now button if unpaid. cpayment.php Card payment UI. On submit sets fine.paid = 1 in database. csignalling.php Read-only view of signal timings (road name, time_from, time_to). croad.php Read-only view of all roads (station name, road name).

ix) 4.4 Officer Pages (prefix: o)

x) osignup.php Registration form for officers. Stores username, email, OID, password. ologin.php Login form. Validates against 'officer' table, sets $_SESSION['email']. ofunctions.php check_login() function - validates session, redirects to ologin.php if absent. ohome.php Officer dashboard with slider linking to all officer features. ofine.php Lists ALL fines in the system with an Edit button per row. ozfupdate.php Edit form to update a specific fine amount by fine ID. osignalling.php Lists all signal timings with an Edit button per row. osupdate.php Edit form to update time_from and time_to for a signal entry. oroad.php View all roads - Road ID, Station Name, Road Name.

xi) Database Schema

xii) Database name: 'project' | Character set: utf8mb4 | Engine: InnoDB

xiii) Table: citizen - Stores registered citizens

xiv) Column Type Constraints

xv) id INT AUTO_INCREMENT, PRIMARY KEY

xvi) username VARCHAR(100) NOT NULL

xvii) email VARCHAR(100) NOT NULL, UNIQUE

xviii) Phone VARCHAR

xix) VID VARCHAR(20) Vehicle ID

xx) pass VARCHAR(255) NOT NULL (plain text - upgrade to hashed in production)

xxi) Table: officer - Stores traffic officers

xxii) OID VARCHAR(20) Officer ID

xxiii) pass VARCHAR(255) NOT NULL

xxiv) Table: fine - Traffic violation fines

xxv) username VARCHAR(100) Links to citizen.username

xxvi) amount DECIMAL(10,2) Default 0.00

xxvii) paid TINYINT 0 = Pending, 1 = Paid

xxviii) Table: road - Road information

xxix) station_name VARCHAR(100)

xxx) road_name VARCHAR(100)

xxxi) Table: signalling - Traffic signal timings per road

xxxii) time_from VARCHAR(20) e.g. 06:00

xxxiii) time_to VARCHAR(20) e.g. 10:00

xxxiv) Complete Application Flow

xxxv) 6.1 Citizen Flow

xxxvi) Step Page Action

xxxvii) Step 1 Visit ahome.php Public homepage. Click 'Register Now'.

xxxviii) Step 2 asuoption.php Choose 'CITIZEN' to go to csignup.php.

xxxix) Step 3 csignup.php (POST) Fill name, email, phone, vehicle ID, password. Two DB inserts: citizen table + fine table (amount=0).

xl) Step 4 clogin.php (POST) Enter email + password. Session set on success. Redirect to chome.php.

xli) Step 5 chome.php Dashboard. Links to Fine Payment, Signal Timings, Google Maps, Road Info.

xlii) Step 6 cfine.php View own fine. If amount > 0 and unpaid, 'Pay Now' link appears.

xliii) Step 7 cpayment.php Enter card details. On submit: fine.paid = 1 in DB. Redirected back with success message.

xliv) Step 8 logout.php Session destroyed. Redirect to ahome.php.

xlv) 6.2 Officer Flow

xlvi) Step 1 ahome.php Click 'Register Now'.

xlvii) Step 2 asuoption.php Choose 'OFFICER' to go to osignup.php.

xlviii) Step 3 osignup.php (POST) Fill name, email, Officer ID, password. Insert into officer table.

xlix) Step 4 ologin.php (POST) Login. Session set. Redirect to ohome.php.

l) Step 5 ohome.php Dashboard. Links to Fine Allotment, Signal Timings, Google Maps, Road Info.

li) Step 6 ofine.php View ALL citizen fines. Each row has an Edit link.

lii) Step 7 ozfupdate.php Edit fine amount for a specific citizen. Updates fine.amount in DB.

liii) Step 8 osignalling.php View all signal timings. Each row has an Edit link.

liv) Step 9 osupdate.php Update time_from and time_to for a road signal.

lv) Step 10 oroad.php View all road records (read-only).

lvi) Step 11 logout.php Session destroyed. Redirect to ahome.php.

lvii) Authentication & Session Management

Both Citizens and Officers use PHP's native session mechanism

i) Login: $_SESSION['email'] = $user_data['email']; (set on successful login) Guard: Every protected page includes check_login($con) from the role's functions file.

ii) If the session is not set, the user is immediately redirected to the login page.

iii) Logout: unset($_SESSION['email']); header('Location: ahome.php');

Two separate check_login() functions exist

i) cfunctions.php - queries the 'citizen' table, redirects to clogin.php if unauthenticated. ofunctions.php - queries the 'officer' table, redirects to ologin.php if unauthenticated.

ii) Setup & Deployment Guide

iii) Requirements

iv) MAMP (or XAMPP) with Apache + MySQL

v) PHP 8.x

vi) Browser

vii) Installation Steps

viii) Copy project folder to /Applications/MAMP/htdocs/Traffic-Monitoring-System/

ix) Open MAMP and click 'Start Servers' (both Apache and MySQL must be green).

x) Open phpMyAdmin at http://localhost:8888/phpMyAdmin

xi) Create a new database named 'project'.

xii) Select 'project' database -> Import tab -> choose database.sql -> click Go.

xiii) Open the site at http://localhost:8888/Traffic-Monitoring-System/ahome.php

xiv) Database Credentials

xv) Parameter Value

xvi) Host localhost

xvii) Username root

xviii) Password root

xix) Database project

xx) Security Notes & Known Limitations

xxi) Issue Detail

xxii) SQL Injection Queries use raw string interpolation. Use prepared statements (PDO/MySQLi) in production.

xxiii) Plain-text passwords Passwords stored as plain text. Use password_hash() and password_verify() in production.

xxiv) No admin role There is no separate admin login. ahome.php is publicly accessible.

xxv) No input sanitization User input is not sanitised before DB insertion. Add htmlspecialchars/strip_tags.

xxvi) Payment is simulated cpayment.php marks fine as paid but does not connect to a real payment gateway.

xxvii) No CSRF protection Forms have no CSRF tokens. Add tokens to all POST forms in production.

xxviii) Frontend Assets & Directory Layout

xxix) Directory Contents

xxx) assets/css/ Bootstrap, Font Awesome, Owl Carousel, Lightbox, main template CSS assets/js/ jQuery, Bootstrap JS, Owl Carousel, ScrollReveal, Lightbox, custom.js assets/images/ Slider images (a.jpg, b.jpg, c.jpg), icons, team photos

xxxi) vendor/bootstrap/ Bootstrap 4 CSS + JS

xxxii) vendor/animate/ Animate.css for scroll animations

xxxiii) vendor/select2/ Select2 dropdown library

xxxiv) vendor/perfect-scrollbar/ Smooth scrollbar for tables vendor/tilt/ Tilt.js - 3D tilt effect on login images vendor/css-hamburgers/ Hamburger menu animation CSS css/ util.css + main.css - login page layout styles

xxxv) font-awesome-4.7.0/ Icon font library

xxxvi) images/icons/ Favicon and SVG illustrations (citizen.svg, officer.svg, traffic.svg)

KEY DETAILS

1. Admin portal manages user accounts, officer assignments, road data, and signalling configurations.

2. Officer portal logs violators, records fines, and updates real-time road and signal status.

3. Citizen portal supports account registration, violation history lookup, and online fine payment.

4. Automated daily reports summarize traffic violations and flag roads needing infrastructure attention.

5. Built with PHP, MySQL (XAMPP), JavaScript, and CSS with a responsive multi-role front-end.

6. Uses PHP session-based authentication with separate login handlers for citizens and officers, each querying its own MySQL table.

7. MySQL database backs five core tables (citizen, officer, fine, road, signalling); citizen signup automatically inserts a linked fine record by username and Vehicle ID.